

Yiling He (Hè Yíling)

Research Fellow | S2Lab, Department of Computer Science, University College London
yiling-he@ucl.ac.uk · [e0hyl.github.io](https://github.com/e0hyl) · [Google Scholar](#) · [GitHub](#) · [ORCID](#)

RESEARCH INTERESTS

AI-driven software security; interpretable and robust machine learning for malware and vulnerability analysis; large language models for code intelligence; AI safety and trustworthy AI for security-critical systems; concept drift in security systems; model explainability.

EDUCATION

Ph.D., Cyberspace Security, *Zhejiang University* Sep. 2021 – Jun. 2024
College of Computer Science | Supervisors: Prof. Kui Ren (ACM/IEEE Fellow), Prof. Zhan Qin
Awards: Outstanding Graduate of ZJU (2024); ZJU Outstanding PhD Dissertation Funding (2023); ZJU Graduate Student of Five Merits (2022); ZJU Outstanding Graduate Student (2021–22); Higgs Innovation Scholarship (2021)

M.Eng., Cyberspace Security (transferred to Ph.D.), *Zhejiang University* Sep. 2019 – Jun. 2021
College of Computer Science | Supervisors: Prof. Kui Ren, Prof. Zhan Qin
Awards: ZJU Outstanding Graduate Student (2019)

B.Eng., Information Security, *Beijing University of Posts and Telecommunications* Sep. 2015 – Jun. 2019
School of Cyberspace Security
Awards: Beijing Outstanding Graduation Design (2019); First Prize, National College Student Information Security Contest (2018); China National Scholarship (2018); Rui'an Tech Scholarship (2017)

ACADEMIC & RESEARCH POSITIONS

Research Fellow, *University College London (UCL), S2Lab* Jan. 2025 – Jun. 2026
Department of Computer Science | Supervisor: Prof. Lorenzo Cavallaro
Research on LLM-powered software security: Android malware analysis, code LLM benchmarking, adversarial robustness of code models.

Research Intern, *Tencent Xuanwu Lab (Intelligent Security Research Group)* Mar. 2021 – Aug. 2021
Contributed to BrutePrint: brute-force attack against smartphone fingerprint authentication ([arXiv:2305.10791](#)).

PUBLICATIONS

Note: **Bold** = Yiling He; [†] = corresponding author. “Sec. Top 4” = CCS/IEEE S&P/USENIX Security/NDSS (the four flagship venues in computer security). Citations as of May 2026 (Google Scholar).

2026

Mingjie Chen, Tiancheng Zhu, Mingxue Zhang, **Yiling He**, Minghao Lin, Penghui Li, and Kui Ren. *Unveiling the Fragility of Binary Code Similarity Detection via Targeted Attacks with Model Explanations*. In *Proc. 34th ACM Int'l Conference on the Foundations of Software Engineering (FSE)*, 2026. [ACM FSE 2026] [\[link\]](#)

Hongwei Yao, Yiming Liu, **Yiling He**[†], and Bingrun Yang. *Red-Teaming Agent Execution Contexts: Open-World Security Evaluation on OpenClaw*. In *2nd Workshop on Agents in the Wild: Safety, Security, and Beyond (ICML 2026 Workshop AIWILD)*, 2026. [ICML 2026 Workshop] [\[PDF\]](#) [\[Code\]](#)

2025

Yiling He, Junchi Lei, Zhan Qin, Kui Ren, and Chun Chen. *Combating Concept Drift with Explanatory Detection and Adaptation for Android Malware Classification (DREAM)*. In *Proc. 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2025. [Sec. Top 4 · 16 citations] [\[DOI\]](#) [\[PDF\]](#) [\[Code\]](#)

Yiling He, Hongyu She, Xingzhi Qian, Xinran Zheng, Zhuo Chen, Zhan Qin, and Lorenzo Cavallaro. *On Benchmarking Code LLMs for Android Malware Analysis (CAMA)*. In *Proc. 34th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*, 2025. [ACM ISSTA 2025 · 23 citations] [\[DOI\]](#) [\[PDF\]](#) [\[Code\]](#) [\[Data\]](#)

Xingzhi Qian, Xinran Zheng, **Yiling He**[†], Shuo Yang, and Lorenzo Cavallaro. *LAMD: Context-driven Android Malware Detection and Classification with LLMs*. In *2025 IEEE Security and Privacy Workshops (SPW)*, 2025. [Best Paper Award · IEEE S&P Workshops 2025 · 58 citations] [\[DOI\]](#) [\[PDF\]](#) [\[Data\]](#)

Shuo Shao, Yiming Li, Hongwei Yao, **Yiling He**, Zhan Qin, and Kui Ren. *Explanation as a Watermark: Towards Harmless and Multi-bit Model Ownership Verification via Watermarking Feature Attribution (EaaW)*. In *Network and Distributed System Security Symposium (NDSS)*, 2025. [Sec. Top 4 · 64 citations] [\[PDF\]](#) [\[Code\]](#)

Xinyu She, Yue Liu, Yanjie Zhao, **Yiling He**, Li Li, Chakkrit Tantithamthavorn, Zhan Qin, and Haoyu Wang. *Pitfalls in Language Models for Code Intelligence: A Taxonomy and Survey*. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2025. [ACM TOSEM · 33 citations] [\[DOI\]](#) [\[PDF\]](#)

Mingzhe Gao, Ligeng Chen, **Yiling He**[†], Yuhang Chen, Lingyun Ying, and Wang Yang. *Distilling Benign Knowledge with Fine-Grained AST Fragments for Precise Real-World Web Shell Detection (ASTBAR)*. In *2025 IEEE/ACM 33rd Int'l Symposium on Quality of Service (IWQoS)*, 2025. [IEEE/ACM IWQoS 2025] [\[DOI\]](#) [\[Code\]](#)

2024

Mengda Xie, **Yiling He**[†], Zhan Qin, and Meie Fang. *RetouchUAA: Unconstrained Adversarial Attack via Realistic Image Retouching*. *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 2024. [IEEE TCSVT · 8 citations] [\[DOI\]](#) [\[PDF\]](#)

2023

Yiling He, Jian Lou, Zhan Qin, and Kui Ren. *FINER: Enhancing State-of-the-art Classifiers with Feature Attribution to Facilitate Security Analysis*. In *Proc. 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2023. [Sec. Top 4 · 22 citations] [\[DOI\]](#) [\[PDF\]](#) [\[Code\]](#)

Zhuo Chen, Jie Liu, Yubo Hu, Lei Wu, Yajin Zhou, **Yiling He**, Xianhao Liao, Ke Wang, Jinku Li, and Zhan Qin. *DeUEDroid: Detecting Underground Economy Apps Based on UTG Similarity*. In *Proc. 32nd ACM SIGSOFT Int'l Symposium on Software Testing and Analysis (ISSTA)*, 2023. [ACM ISSTA 2023 · 14 citations] [\[DOI\]](#) [\[PDF\]](#)

2022

Yiling He, Yiping Liu, Lei Wu, Ziqi Yang, Kui Ren, and Zhan Qin. *MsDroid: Identifying Malicious Snippets for Android Malware Detection*. *IEEE Transactions on Dependable and Secure Computing (TDSC)* 20(3): 2025–2039, 2022. [IEEE TDSC · 91 citations] [\[DOI\]](#) [\[Code\]](#)

Under Review / Conditionally Accepted

Yiling He, Junchi Lei, Hongyu She, Shuo Shao, Xinran Zheng, Yiping Liu, Zhan Qin, and Lorenzo Cavallaro. *Retrofit: Continual Learning with Controlled Forgetting for Binary Security Detection and Analysis*. *USENIX Security Symposium 2026* (Shepherd Accepted, conditional on artifact evaluation). [Sec. Top 4] [\[PDF\]](#) [\[Code\]](#)

Xinran Zheng, Xingzhi Qian, **Yiling He**[†], Shuo Yang, and Lorenzo Cavallaro. *Is “Knowing It’s Malicious” Enough? Evaluating LLMs for Fine-Grained Malware Behavior Auditing*. *ACM ISSTA 2026* (under major revision).

Yuchen Yang, Yiming Li, Hongwei Yao, Bingrun Yang, **Yiling He**, Tianwei Zhang, Dacheng Tao, and Zhan Qin. *ShadowCode: Towards (Automatic) External Prompt Injection Attack against Code LLMs*. *IEEE TDSC* (under minor revision). [12 citations] [\[PDF\]](#)

TALKS & PRESENTATIONS

- **Berkeley Security Seminar**, UC Berkeley (Soda Hall, Rm. 380), Oct. 1, 2025.
Combating Concept Drift with Explanatory Detection and Adaptation for Android Malware Classification (invited talk). [\[link\]](#)
- **ACM CCS 2025**, Taipei, Taiwan, Oct. 13–17, 2025 (Session B7).
DREAM: Combating Concept Drift with Explanatory Detection and Adaptation (paper presentation).
- **LLMSC 2025 Workshop** (co-located with ISSTA 2025), Trondheim, Norway, Jun. 28, 2025.
On Benchmarking Code LLMs for Android Malware Analysis (paper presentation, 09:45–10:00). [\[program\]](#)
- **ACM CCS 2023**, Copenhagen, Denmark, Nov. 26–30, 2023.
FINER: Enhancing State-of-the-art Classifiers with Feature Attribution (paper presentation).
- **Qatar Computing Research Institute (QCRI), HBKU**, Doha, Qatar, Nov. 23, 2023.
FINER: Enhancing State-of-the-art Classifiers with Feature Attribution to Facilitate Security Analysis (invited talk, extended version).
- **HUST Security Pride Research Group Seminar**, Jun. 18, 2024 (15:00–16:00).
Advancing Explainability and Adaptability in AI-based Software Security (invited talk).

AWARDS & HONORS

Best Paper Award, IEEE Security and Privacy Workshops (LAMD)	2025
Outstanding Graduate, Zhejiang University	2024
ZJU Outstanding PhD Dissertation Funding Recipient	2023
ZJU Graduate Student of Five Merits	2022
Higgs University Student Extracurricular Innovation Scholarship	2021
ZJU Outstanding Graduate Student	2019, 2021–22
Beijing Outstanding Graduation Design (city-level)	2019
First Prize, National College Student Information Security Contest (national)	2018
China National Scholarship (national, Ministry of Education)	2018
Rui’an Tech Scholarship	2017

PROFESSIONAL SERVICE

Programme Committee Member

- ACM SIGSAC Conference on Computer and Communications Security (CCS) 2026
- IEEE Symposium on Security and Privacy (IEEE S&P) 2026
- ACM Workshop on Artificial Intelligence and Security (AISec, co-located with CCS) 2025
- Workshop on Rethinking Malware Analysis (WoRMA, co-located with IEEE EuroS&P) 2025

External Reviewer

- ACM CCS 2022, 2024, 2025
- IEEE Symposium on Security and Privacy (IEEE S&P) 2025
- USENIX Security Symposium 2025

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC)
- ACM Transactions on Privacy and Security (TOPS)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- Tsinghua Science and Technology (TST)
- Int'l Journal of Software Engineering and Knowledge Engineering (IJSEKE)

CITATION METRICS

Retrieved from Google Scholar on May 29, 2026.

Total citations	359
<i>h</i> -index	10
<i>i</i> 10-index	10

scholar.google.com/citations?user=l55RAe8AAAAJ

OPEN-SOURCE SOFTWARE

Repository	Description	Stars
MsDroid	Android malware detection via GNN-based malicious-snippet identification	67
AndrozooDownloader	High-performance AndroZoo dataset downloader	57
BlockImage_System	Blockchain-based image sharing with digital watermarking	33
DREAM-drift-adapt	Concept-drift detection and adaptation for Android malware classifiers	22
CallGraph-Flowdroid	Call-graph extraction from Android APKs via FlowDroid	20
FINER-explain	Explanation framework for risk-detection classifiers	12